

Disruptions in Art, Science, Engineering and Modeling

SIGGRAPH 2022 Retrospective Panel

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ABSTRACT

An overview of the ACM SIGGRAPH Retrospectives panel on the impact of computer graphics.

KEYWORDS

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1 INTRODUCTION

We present a one hour panel from four computer graphics luminaries. The goal of the panel is to inspire the next generation of attendees to the SIGGRAPH conferences. To help those on a career path in computer graphics, whether at the beginning or seasoned researchers. There are many pathways for a career in computer graphics to have real impact. We present some untold stories and hear from influential voices in the field. Our panelists are innovators and disruptors. We look at how these researchers, using CG have caused the world to change, in art, science, engineering, and modeling. We present their pathways to success, as an inspiration for young SIGGRAPHers to strive towards making disruptions in their own chosen fields.

2 THE PANEL

Professor Brian Wyvill (Moderator) is a Canadian computer scientist and author who is currently a professor emeritus at the University of Victoria. He was a vice-president of ACM SIGGRAPH from 2013 to 2020. He is regarded as a pioneer in the field of implicit modelling for computer graphics and animation. His most prominent



Figure 1: Image from *Sketch-Based Implicit Blending* Baptiste Angels et al SIGGRAPH Asia 2017

contributions include developing the first implicit surface polygonizer with his brother Geoff Wyvill and the BlobTree, a single unified structure for representing both the topology and geometry of complex implicit models. Inspired by the work of panelist, Alyn Rockwood, he also challenges the conventional reliance on polygons. He has published more than 150 peer-reviewed articles, several books including one novel, and shown two animated movies at the SIGGRAPH Electronic Theatre.

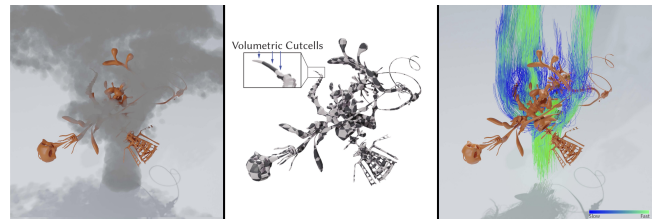


Figure 2: Fluid flow around complex shapes - VEMPIC: Particle-in-Polyhedron Fluid Simulation for Intricate Solid Boundaries, SIGGRAPH 2022

Professor Eugene Fiume (Science Panelist) is Professor and Dean of the Faculty of Applied Sciences at Simon Fraser University. Eugene's work is distinguished by formal, mathematical frameworks through which a deeper understanding of objects and their behaviours to support more efficient and accurate rendering. He co-developed symmetric orthogonal spherical wavelets and collaborated on the introduction of reproducing kernel Hilbert spaces

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to computer graphics. He has made contributions to biomedical computing and high-performance graphics. He has published two books and over 130 research papers. He holds innovation Awards from ITRC, Burroughs-Welcome, an NSERC Synergy Award for innovation and Fellow of the Royal Society of Canada. He was papers chair for SIGGRAPH 2001. Eugene disrupts by taking ad hoc methods and defining rigid mathematical and scientific principles to achieve advancements on a practical level.

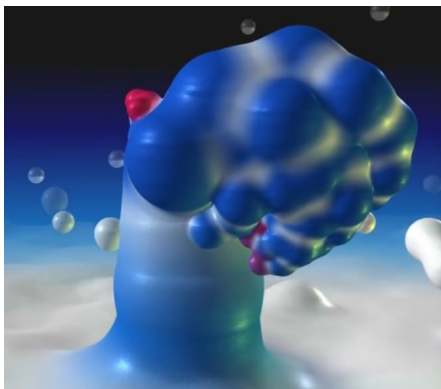


Figure 3: An Image from *Growth : Mysterious Galaxy* presented at the ACM SIGGRAPH Electronic Theater in Detroit 1983.

Professor Yoichiro Kawaguchi (art panelist) produced revolutionary art work based on metaballs in the early 1980s, a form of implicit modeling. These organic images and animations were ray traced on the first purpose built ray tracing computer called the Links-1 with Professor Koichi Omura at the University of Osaka. This work was an early example of art as part of the artificial life movement. Kawaguchi's entire career has been dedicated to innovation in this area. He has pushed the use of mathematics and algorithms to disrupt the world of art. He taught at the Nippon Electronics College, and the University of Tokyo where he introduced the cross-disciplinary teaching of fine art and technology. His contributions have helped make these tools a serious part of the art's fabric in the 21st century. His contributions to SIGGRAPH over the last 35 years are legendary, including the annual sake barrel.

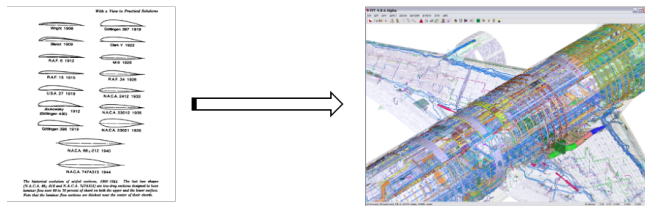


Figure 4: Evolution of Aerospace Design

Dr. Dave Kasik (Engineering Panelist) retired as Boeing's Senior Technical Fellow in visualization and interactive techniques in 2016. He advanced Boeing's use of digital twin 3D design and digital thread processes. Dave devoted his first 11 years at Boeing to research and development of computer-aided design software and led

to pioneering work in interactive 3D graphics, user interface management systems, and industrial use of NURBS solids and surfaces. Later projects introduced Boeing to Massive Model Visualization and visual analytics. He is also a SIGGRAPH supporter, introducing new programs, service to the pioneers and driven the support for high schoolers to become exposed to the wonders of SIGGRAPH.

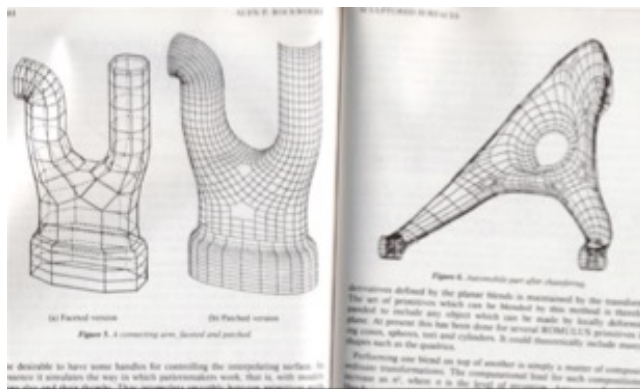


Figure 5: Images from the first commercial Recursive Subdivision software. Uses the Sabin/Doo scheme. REMUS™ from Shape Data Ltd. 1983

Dr. Alyn Rockwood (Modeling Panelist) is a retired Professor of mathematics, and Associate Director of research at KAUST in Saudi Arabia. He has been conducting research in computer graphics for 40 years, including the first certification for a pilot training simulator at Evans and Sutherland, and a rendering algorithm for curved surfaces at SGI for OpenGL. For SIGGRAPH, he was papers chair in 1999 (SIG Asia in 2013) and conference chair in 2003. His awards include COFES 2007 Innovation in Technology, CAD Society "Heroes of Engineering", and SIGGRAPH's Outstanding Service. Publications include over 100 peer-reviewed articles, 10 patents, and 4 books, including 2 novels. He received his Ph.D. from the University of Cambridge, UK. He has introduced disruptions to CG through new modeling techniques, pioneering advanced ideas in mathematical modeling. Some of his early innovations are only now coming to fruition as computers become fast enough to keep up with Alyn's ideas.

3 QUESTIONS FOR THE PANEL

The panelists gave a personal view in answer to four questions put by the organizer. These were the following:

- (1) Describe the path you took to have an impact?
- (2) What role did SIGGRAPH play in helping your career?
- (3) What was your most pleasing innovation ?
- (4) What happened in your work that surprised and/or delighted you?

The answers to these questions varied considerably, presenting lessons learned from each participant. The anecdotes told were amusing, informative and surprising.